

SAFI, Fardin Abrar

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Education

The Hong Kong University of Science and Technology, HKSAR

Expected Graduation: August 2028

B.Eng. in Computer Engineering

HKSAR Government Scholar | HKUST University Scholar | Engineering Student Ambassador | Computer Engineering Mentor

Research and Professional Experience

The Hong Kong University of Science and Technology

May 2026 – Aug 2026

Undergraduate Researcher – AI in Biomedicine

- Conducted research on improving Continuing Medical Education (CME) using generative AI and data analytics. Developed an LLM-powered learning module based on Miller's Pyramid to enhance learner retention beyond traditional CME approaches, and designed pilot studies to assess effectiveness and identify improvement opportunities.

Oxbridge Economics, Cyberport, HK

May 2026 – Jul 2026

Data Scientist

- Developed US Central Banks module for FinAnalytics (FinFAST.ai's AI-powered econometric & macro data platform), expanding coverage from China to the US via web scraping and data acquisition of Federal Reserve statements and FOMC transcripts.
- Built end-to-end automated data pipeline (MongoDB schema design, scheduled scraper, ETL processing, data cleansing and standardization) and migrated frontend from Vite to Next.js with full platform integration.
- Improved structured data extraction accuracy from 59% to 85% using NLP, refined LLM prompts and generative AI techniques; implemented sentiment analysis on FOMC notes to produce quantitative scores for econometric modeling and risk identification.

Hong Kong Hospital Authority, Kwun Tong, HK

Jul 2025 – Jun 2026

Artificial Intelligence Engineer

- Architected a Python-based Playwright data-ingestion and data acquisition pipeline to bypass source-code and API limitations of a closed vendor chatbot, enabling real-time response extraction and data verification.
- Engineered a low-latency, multilingual (English, Mandarin, Cantonese) interactive avatar interface utilizing speech-to-text and text-to-speech (TTS) streaming pipelines.
- Integrated Azure TTS alongside Windows-native TTS overrides to fill regional language support gaps, implementing audio chunking strategies to minimize response latency.
- Conducted architectural evaluations comparing RAG versus fine-tuning methodologies across MedGemma, Qwen, and DeepSeek models, establishing a production-ready RAG pipeline with 90%+ accuracy for clinical guideline retrieval and data quality assurance.
- Built layout-aware PDF extraction pipelines using open-source parsers (MinerU) to ingest complex medical tables and images; mitigated hallucination and overfitting via few-shot "admittance of ignorance" prompting and LLM-synthesized QA training pairs for robust data standardization.
- Architected an end-to-end document processing and data analytics pipeline using Gradio, leveraging Qwen-VL to parse complex layouts from PDFs and Excel files into clean, structured Markdown formats supporting downstream analysis and visualization.
- Implemented page-level API chunking methodologies to optimize pipeline throughput, orchestrating downstream LLMs and custom Python deduplication scripts to generate exhaustive, non-repetitive clinical assessment engines.
- Automated complex data import/export, cleansing, and document workflows using Python scripts, reducing manual processing time by 60% while maintaining data integrity and compliance considerations.

TabNext IT Healthcare, Lai Chi Kok, HK

Jun 2025 – Jun 2025

Front-End Programmer

- Engineered a custom, open-source-grade Vue.js calendar component to resolve multi-slot rendering fragmentation, developing a unified time-blocking algorithm that consolidated split data intervals into continuous UI blocks for improved data visualization.
- Architected modular UI adaptation layers to rapidly customize the core Clinical Management System (CMS) for diverse client verticals (e.g., medical clinics, veterinary facilities) during competitive sales pitches, supporting client analytics needs.

Igears Technology Ltd., Kwai Fong, HK

Jun 2024 – Aug 2024

Full Stack Programmer

- Engineered a client-facing web application that aggregates and benchmarks network telemetry data, integrating multiple third-party speed-test APIs to provide real-time, comparative performance analytics and data-driven insights.
- Designed dynamic data visualization dashboards for cross-service analytics and managed the end-to-end deployment, configuration, and maintenance of the production application on enterprise Linux servers.

Hackathons

SuperEd: AI-Powered Personalized Education Platform

IBM x Watson Hackathon – Bronze Medal | Oct 2025

- Led the full-stack development of an adaptive university learning companion using Next.js, MongoDB, and IBM Watsonx LLMs (generative AI) to map real-time student comprehension profiles and support data-driven insights.
- Engineered dynamic prompting topologies and stateful backend architectures to automatically generate evolving personalized learning graphs and predictive future study paths through advanced analytics.

Skills

Data Analytics & Engineering: Data Analytics, Data Acquisition, Data Cleansing, Data Standardization, ETL Pipelines, Data Visualization, Dashboards, Database Design (MongoDB), Data Quality Assurance, Statistical Analysis, Risk Identification, Due Diligence Support

AI/ML, Tools & Prompt Engineering: Generative AI, RAG, Vision-Language Models (Qwen-VL), LLM Fine-Tuning (MedGemma, DeepSeek), NLP, Sentiment Analysis, Agentic Coding (Claude Code, GitHub Copilot Workspace), Advanced Prompting, Synthetic Data, Multimodal Pipelines, Python, SQL

Languages: English (Fluent); Bengali, Hindi, Urdu (Native); French (Intermediate); Cantonese, Mandarin (Conversational) | **Other:** Microsoft Office, Analytical Problem-Solving, Client Service Orientation, Collaboration